

Machine Learning Journey

Welcome to my **Machine Learning Journey** 🚀

This repository contains my learning progress, notes, experiments, and projects as I learn Machine Learning step by step.

The goal of this repository is to build a strong understanding of **Machine Learning fundamentals**, from mathematics and statistics to model building, evaluation, and eventually real-world ML projects.

What I'm Learning

1. Python for Machine Learning

- Python basics
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Data manipulation and visualization

2. Exploratory Data Analysis (EDA)

- Understanding datasets
- Handling missing values
- Detecting outliers
- Understanding distributions
- Correlation analysis
- Data visualization

3. Statistics

- Mean, Median, Mode
- Variance and Standard Deviation
- Z-score
- Probability basics
- Statistical concepts used in Machine Learning

4. Supervised Learning

Regression

- Linear Regression
- Cost Function
- Gradient Descent

- Ridge Regression
- Lasso Regression
- Regularization
- Model evaluation

● Classification

- Logistic Regression
- Classification concepts
- Decision boundaries
- Classification performance metrics

5. Model Evaluation

Currently learning:

- R^2 Score
- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- Cross-Validation

Current Projects / Notebooks

Linear Regression, Ridge and Lasso

This notebook covers:

- Linear Regression
- Finding patterns in data
- Cost function
- Gradient Descent
- Parameter optimization
- Ridge Regression
- Lasso Regression
- Regularization
- Model performance evaluation

More notebooks and projects will be added as I continue learning.

Tools & Technologies

-  Python
 -  NumPy
 -  Pandas
 -  Matplotlib
 -  Seaborn
 -  Scikit-learn
 -  Jupyter Notebook
 -  Git
 -  GitHub
-

Learning Roadmap

My current learning path:

```
Python
↓
NumPy & Pandas
↓
Data Visualization
↓
EDA
↓
Statistics
↓
Linear Regression
↓
Gradient Descent
↓
Ridge & Lasso
↓
Classification
↓
Model Evaluation
↓
Cross-Validation
↓
Feature Engineering
↓
More Machine Learning Algorithms
↓
Projects
↓
Deployment
```

Progress

This repository will be updated continuously as I learn new concepts and build projects.

Topic	Status
Python Basics	✓
NumPy	✓
Pandas	✓
Matplotlib	✓
Seaborn	✓
Basic EDA	✓
Basic Statistics	↺↻
Linear Regression	✓
Gradient Descent	✓
Ridge Regression	✓
Lasso Regression	✓
Logistic Regression	↺↻
Model Evaluation	↺↻
Cross-Validation	↺↻
Feature Engineering	🕒
Advanced ML	🕒
ML Projects	🕒
Deployment	🕒
MLOps	🕒

Legend:

✓ Completed | ↺↻ Currently Learning | 🕒 Coming Next

Goal

My long-term goal is to develop strong practical skills in **Machine Learning and AI Engineering** by:

- Understanding the theory behind algorithms
 - Implementing concepts using Python
 - Working with real datasets
 - Building end-to-end projects
 - Learning model deployment
 - Understanding production ML workflows
 - Continuously improving through projects and experimentation
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Git Workflow

I use Git and GitHub to track my learning progress.

```
git add .  
git commit -m "Describe your changes"  
git push
```

This repository will serve as a record of my progress throughout my Machine Learning journey.

Follow My Progress

This repository is a work in progress.

New concepts, notebooks, experiments, and projects will be added as I continue learning.

Learning → Building → Experimenting → Improving

★ Thanks for visiting my Machine Learning Journey!